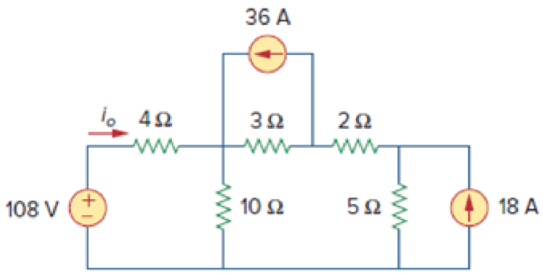
Chapter 4, Problem 16P

Problem

Given the circuit in Fig. 4.84, use superposition to obtain *io*.

Figure. 4.84:



Step-by-step solution

Step 1 of 9

Refer to Figure 4.84 in the textbook.

Apply superposition theorem to the circuit and calculate the individual currents for the individual sources acting alone. There are three independent sources in the circuit. Assume i_1 is the current due to 108 V voltage source, i_2 is the current due to 36 A current source and i_3 is the current due to 18 A current source. Therefore,

$$i_o = i_1 + i_2 + i_3$$

Consider 108 V voltage source and replace 36 A and 18 A current sources with open circuit to find the current i_1. Draw the modified circuit diagram.

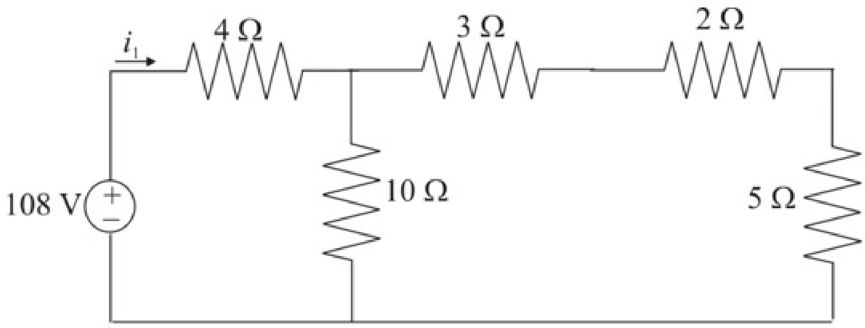


Figure 1

Step 2 of 9

In Figure 1, the series combination of resistors $3\ \Omega, 2\ \Omega$ and $5\ \Omega$ is in parallel with $10\ \Omega$ resistor. Now this entire combination is in series with $4\ \Omega$ resistor. Determine equivalent resistance of the circuit.

$$\begin{aligned} R_{eq} &= 4+(10\,\|\,(3+2+5)) \\ &= 4+(10\,\|\,10) \\ &= 4+\left(\frac{10\times 10}{10+10}\right) \\ &= 4+5 \\ &= 9\ \Omega \end{aligned}$$

Determine the value of current i_1 using Ohms law.

$$\begin{aligned} i_1 &= \frac{108}{R_{eq}} \\ &= \frac{108}{9} \\ &= 12\ \text{A} \end{aligned}$$

Therefore, the value of the current i_1 due to 108 V voltage source is $12\ \text{A}$.

Step 3 of 9

Consider 36 A current source and replace 108 V voltage source with short circuit and 18 A current source with open circuit to find the current i_2. Draw the modified circuit diagram.

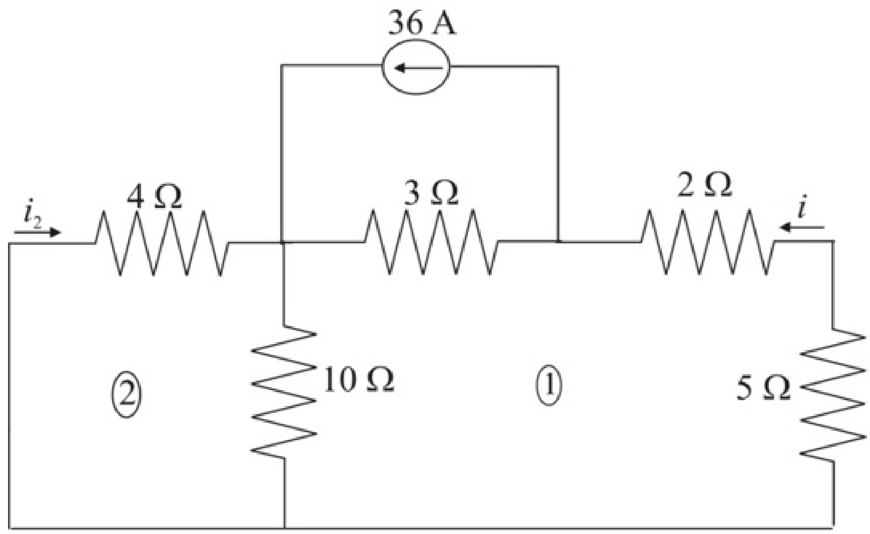


Figure 2

Step 4 of 9

Apply Kirchhoff's Voltage Law in loop 2.

$$\begin{aligned} &4i_2+10(i+i_2)=0 \\ &4i_2+10i+10i_2=0 \\ &14i_2+10i=0 \\ &10i=-14i_2 \\ &i=-1.4i_2 \end{aligned}$$

Step 5 of 9

Apply Kirchhoff's Voltage Law in loop 1.

$$5i+2i+3(i-36)+10(i+i_2)=0$$
$$7i+3i-108+10i+10i_2=0$$
$$20i-108+10i_2=0$$

Substitute $-1.4i_2$ for i.

$$20(-1.4i_2)-108+10i_2=0$$
$$-28i_2+10i_2=108$$
$$-18i_2=108$$
$$i_2=-\frac{108}{18}$$
$$i_2=-6\text{ A}$$

Therefore, the value of the current i_2 due to 36 A current source is -6 A.

Step 6 of 9

Consider 18 A current source and replace 108 V voltage source with short circuit and 36 A current source with open circuit to find the current i_3. Draw the modified circuit diagram.

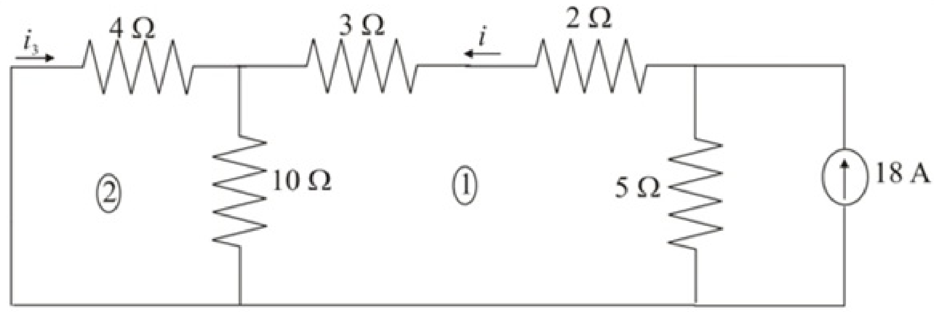


Figure 3

Step 7 of 9

Apply Kirchhoff's Voltage Law in loop 2.

$$4i_3+10(i+i_3)=0$$
$$4i_3+10i+10i_3=0$$
$$14i_3+10i=0$$
$$10i=-14i_3$$
$$i=-1.4i_3$$

Step 8 of 9

Apply Kirchhoff's Voltage Law in loop 1.

$$5(i-18)+2i+3i+10(i+i_3)=0$$
$$5i-90+5i+10i+10i_3=0$$
$$20i-90+10i_3=0$$

Substitute $-1.4i_3$ for i.

$$20(-1.4i_3)-90+10i_3=0$$
$$-28i_3+10i_3=90$$
$$-18i_3=90$$
$$i_3=-\frac{90}{18}$$
$$i_3=-5\text{ A}$$

Therefore, the value of the current i_3 due to 18 A current source is -5 A.

Step 9 of 9

Calculate the total current, i_o when all sources are acting together.

$$i_o=i_1+i_2+i_3$$

Substitute 12 A for i_1, -6 A for i_2 and -5 A for i_3.

$$\begin{aligned}i_o&=12-6-5\\&=1\text{ A}\end{aligned}$$

Therefore, the value of the current i_o in the circuit is $\boxed{1\text{ A}}$.